SOFTWARE ENGINEERING LAB - LAB 3: WRITTEN JUSTIFICATION

Architectural Pattern Selection for Self-Service Coffee Kiosk System

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Architecture Selection: "We chose Layered Architecture for the Self-Service Coffee Kiosk System."

1. Architectural Choice

We selected a 3-Tier Layered Architecture structured into a Presentation Layer (Touchscreen GUI), a Business Logic Layer (Order Manager Component and Payment Service Component), and a Hardware Abstraction & Persistence Layer (Receipt Printer Driver, EMV Card Reader Interface, and Menu & Pricing Service). For an on-premise, single-physical-device kiosk operating within a café, Layered Architecture provides the ideal balance of modularity, low maintenance complexity, and direct hardware controllability without the distributed networking overhead of Microservices or the single-point-of-failure vulnerabilities of Client-Server architectures.

2. Two Scenario-Related Reasons for Selection

• **Reason 1 — Hardware Abstraction & Peripheral Modularity:** The kiosk relies heavily on dedicated physical peripherals: a touch screen monitor, an ESC/POS thermal receipt printer, and an EMV credit card reader. The Layered pattern isolates device drivers into the lowest tier behind clean interfaces (IReceiptPrinter and ICardReaderHardware). This completely decouples higher-level business logic (Order Manager) from vendor-specific serial/USB communication protocols. If the receipt printer runs out of paper or is replaced with a different hardware model, only the hardware driver tier is affected.

• **Reason 2 — Autonomous Edge Resilience & Offline Availability:** A busy café environment requires continuous, uninterrupted operation regardless of transient Wi-Fi drops or external network outages. Unlike a thin Client-Server model that halts ordering when backend connections fail, the Layered kiosk operates autonomously on local hardware. Menu browsing (Espresso, Americano, Latte in Small/Large), price calculations, and receipt formatting continue executing locally with zero latency and 100% kiosk uptime.

3. Security Advantage (PCI-DSS Compliance via Layered Isolation)

Because credit card is the exclusive payment method, strict PCI-DSS data protection is required. The Layered Architecture establishes a rigorous security boundary by encapsulating the Payment Service Component and EMV card hardware away from the Presentation Layer and Order Manager. Cardholder track data is processed and encrypted at the hardware/payment layer and is never transmitted to or logged in general memory. The Order Manager receives only sanitized authorization tokens through the narrow IPaymentProcessor interface, minimizing the system's attack surface and reducing audit scope.

4. Performance Benefit (Zero-Latency In-Memory Execution)

In a busy café with peak morning queues, customer turnaround time directly affects sales and satisfaction. Layered Architecture eliminates inter-service network serialization, TCP handshakes, and remote JSON parsing overhead inherent to Microservices. Inter-component method calls (Touchscreen UI → Order Manager → Menu Service) execute via sub-millisecond in-process invocations (< 2 ms). Rapid in-memory lookup of menu items and pricing ensures instantaneous 60 fps touch responsiveness, and receipts are dispatched to the printer immediately upon payment confirmation.